Symptom:

P0509-IAC VALVE SIGNAL CIRCUIT HIGH

When Monitored and Set Condition:

P0509-IAC VALVE SIGNAL CIRCUIT HIGH

When Monitored: Engine running. Battery voltage greater than 10 volts. IAC motor operating.

Set Condition: The PCM senses a short to ground or battery voltage on the Linear Idle Air Control (LIAC) control circuit for 2.75 seconds while the IAC motor is active.

POSSIBLE CAUSES
IAC MOTOR OPERATION
IAC MOTOR
(K961) IAC SIGNAL CIRCUIT SHORTED TO VOLTAGE
(K61) IAC CONTROL CIRCUIT SHORTED TO VOLTAGE
(K61) IAC CONTROL CIRCUIT SHORTED TO (K961) IAC SIGNAL CIRCUIT
PCM

TEST	ACTION	APPLICABILITY
1	**NOTE: If the engine will not idle, maintain an engine speed between 800 and 1500 RPM.** Start the engine. Allow the engine to idle. Check for vacuum leaks, proper PCV valve installation, proper MAP Sensor installation, and proper IAC installation. With the DRBIII®, read the IAC Current. Is the IAC Current above 999 mA? Yes → Go To 2 No → Refer to the INTERMITTENT CONDITON Symptom (Diagnostic Procedure). Perform POWERTRAIN VERIFICATION TEST VER - 5.	All
2	Turn the ignition off. Disconnect the IAC Motor harness connector. With the DRBIII®, monitor the IAC Current. Ignition on, engine not running. Does the DRBIII® display IAC Current at 0mA? Yes → Replace the IAC Motor. Perform POWERTRAIN VERIFICATION TEST VER - 5. No → Go To 3	All